Theo GUYARD

French nationality | 21 years old | B driving licence

I am currently looking for a 4-month end-of-study internship starting in mid-February 2021 (or later) and a PhD. I'd like to find an internship in a company that would also offer me a CIFRE PhD. I would like to find a subject in applied mathematics that would allow me to put my theoretical work into practice.

Contact

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Rennes (35000), France

Diplomas

- Cambridge English Certificate (B1)
- French Baccalaureate, «very good» mention with jury's congratulations
- Handimanagment degree from Companieros
- INSA engineer specialized in Mathematics (expected in 2021)

Languages

- French (native language)
- English (970/990 at TOEIC)
- Spanish (B1 level)

Hobbies

- Music (trumpet, piano, jazz)
- Eco-building
- Gardening
- Trekking, Skiing, surfing, biking

Schooling

2018 - 2020 INSA Rennes - Mathematics Department

- Theoretical and practical courses about : Optimization, Operation Research, Statistics, Machine Learning, Data Science, Simulation, Informatic tools, Risk analysis, Stochastic models, Time series.
- Languages : English / Option : Music

2016 - 2018 INSA Rennes - Preparatory class

- Basic tools for engineer courses : Mathematics, Mecanic, Electronic, Physic, Chemistry, Informatic, Automatic, ...
- Languages : English, Spanish / Option : Music

Experiences

2020- 2021 IRMAR - Research program

Robust optimization methods for the European Kidney Exchange Program. Supervized by Jeremy Omer and Ayse Nur Arlsan.

2019 - 2020 INRIA Rennes - Research internship

Fast LASSO solving methods using screening tests. Supervized by Cedric Herzet and Clement Elvira. Paper submission with our new screening tests.

2019 - 2020 INRIA Rennes - Research program

Parsimony, Screening methods, Optimization in high dimension. Supervised by Cedric Herzet and Clement Elvira.

Summer 2019 IRMAR - Research internship

Optimization for a ski slope grooming problem. Operation Research about routing problems. Supervized by Jeremy Omer.

Mathematical skills

Optimization	Convex and non-smooth optimization, Gradient and proximal based methods, ADMM, High-dimentional optimization.
Parcimony	Sparse models, Screening methods, Penalized models (LASSO/Ridge/Elastic-net), PCA.
Op. Research	Linear and integer modelling, Branch-and-Bound/Cut/Price, Graphs, Polyhedron theory, Robust optimization, Cutting planes, Bender's decomposition, Stochastic optimization.
Machine Learning	Neural and Bayesian networks, Bagging, Boosting, Random forest, Binary trees, SVM.
Statistics	Scoring, Mathematical statistics, Linear and logistic regression, Analysis and forecasting, Time series (ARMA/ARIMA).
Num. methods	Linear systems, Interpolation, FFT, Matrix decomposition.
Theoretical maths	Analysis (real and complex), Algebra, Topology.

Informatical skills

Python | R | Matlab | Julia | C / C++ | Web (Django)

Publications

In proceeding Turbo Screening for LASSO - IEEE ICASSP 2021 Toronto

Associative commitment

2019 - 2020 AEIR organization, Board member

- Management of the legal and administrative part of the association.
- ~1000 annual subscribers. Yearly budget : 900k €

2016 - 2021 INSA Rennes and student organizations

- Responsabilities in different clubs of AEIR organization : Environment, Gala, Rock'n Solex, Cafeteria, and more ...
- Elected to the preparatory and mathematical school council at INSA Rennes.